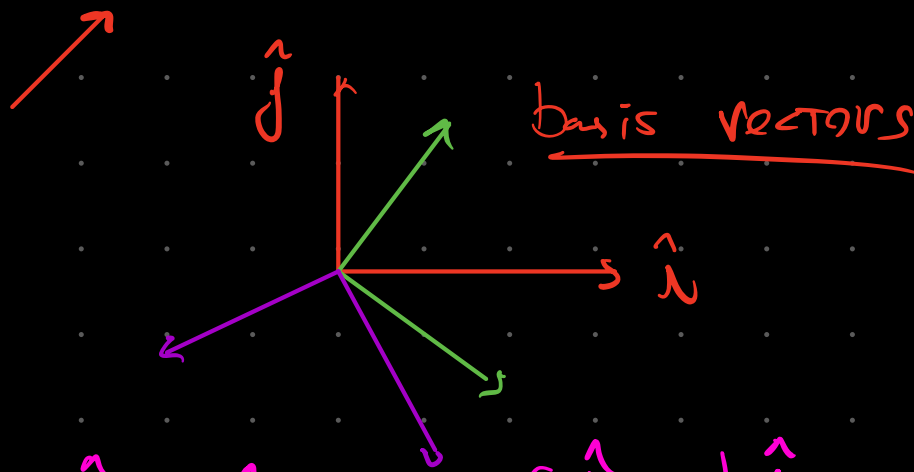


Vector



Span as $\hat{j}$ and $\hat{i}$

$$a\hat{j} + b\hat{i}$$

($\hookrightarrow$ The entire 2D space)

linear combinations

$\hat{i}, \hat{j}, \hat{k} \rightarrow$ span 3D space

Matrices as linear Transforms

keep lines as lines

and not change the origin